kinematics of rigid body

Exercise 01:

1. Represent the points A and B with polar coordinates: $(1,2)$ and $\left(2,\frac{5\pi}{4}\right)$;
2. Represent the points C and D with cylindrical coordinates: $\left(2,\frac{\pi}{2},2\right)$ and $(1,\pi,-1)$.
3. Express the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ in the corresponding local coordinate systems.
4. Represent the points E and F with spherical coordinates: $\left(2,\frac{\pi}{4},\frac{\pi}{4}\right)$ and $\left(1,\frac{\pi}{3},\frac{3\pi}{4}\right)$; Express the vector $\overrightarrow{EF}$ in the local coordinate system $(E,\overrightarrow{e_r},\overrightarrow{e_\theta},\overrightarrow{e_\varphi})$.

Exercise 02:

A material point moves along a trajectory described by the following parametric equations: $\begin{cases} x = t \\ y = 2t^2 \\ z = 0 \end{cases}$

Determine:

1. The unit tangent vector $\vec{u}$ to the trajectory;
2. The radius of curvature ρ
3. The normal $\vec{n}$ to the trajectory;
4. The binormal $\vec{b}$.